THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $6x^2 + 19x + 10$	Multiply term by term and collect like terms
2	**(c)** $4x^2 - 12xy + 9y^2$	$(a-b)^2 = a^2 - 2ab + b^2$
3	**(d)** 2600	$(63+37)(63-37) = 100 \times 26$
4	**(d)** 100	$(7+3)^2$
5	**(d)** 54	$(4+2)(4+5)$
6	**(a)** $8x$	The x^2 and constant terms cancel
7	**(a)** $2x + 22$	Expand each bracket and collect like terms
8	**(c)** $12x^2 + 11x + 2$	Multiply term by term and collect like terms
9	**(c)** $9x^2 - 6xy + y^2$	$(a-b)^2 = a^2 - 2ab + b^2$
10	**(d)** 5600	$(78+22)(78-22) = 100 \times 56$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(2x)^2 + 2(2x)(5y) + (5y)^2$ (1) $= 4x^2 + 20xy + 25y^2$ (1).	1+1
12	$39 \times 41 = (40)^2 - (1)^2$ (1) $= 1600 - 1 = 1599$ (1).	1+1
13	$9x + 27 - 4x + 8$ (1) $= 5x + 35$ (1).	1+1
14	$(3x)^2 + 2(3x)(2y) + (2y)^2$ (1) $= 9x^2 + 12xy + 4y^2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$6x^2 - 15x + 4x - 10$ (2); $= 6x^2 - 11x - 10$ (1).	2+1
16	$197 = 200 - 3$ (1); $(200-3)^2 = 40000 - 1200 + 9$ (1); $= 38809$ (1).	1+1+1
17	LHS $= 49 + 9 = 58$ (1.5); RHS $= 2(25+4) = 58$ (1.5).	1½+1½
	OR: $12x^2 - 8x + 3x - 2$ (2); $= 12x^2 - 5x - 2$ (1).	2+1
18	$108 = 110 - 2$ (1); $(110-2)^2 = 12100 - 440 + 4$ (1); $= 11664$ (1).	1+1+1
	OR: LHS $= 121 + 9 = 130$ (1.5); RHS $= 2(49+16) = 130$ (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Plot $= (3x+2)^2 = 9x^2 + 12x + 4$ (2); pond $= 4x^2$ (1); remaining $= 5x^2 + 12x + 4$ (1); at $x = 10$: 624 m^2 (1).	2+1+1+1
	OR: (i) $103^2 = 10609$ (1.5); (ii) $88 \times 92 = 8096$ (1.5); (iii) $(76+24)(76-24) = 100 \times 52 = 5200$ (2).	1½+1½+2
20	Plot $= (4x+5)^2 = 16x^2 + 40x + 25$ (2); pond $= 4x^2$ (1); remaining $= 12x^2 + 40x + 25$ (1); at $x = 10$: 1625 m^2 (1).	2+1+1+1
	OR: (i) $97^2 = 9409$ (1.5); (ii) $59 \times 61 = 3599$ (1.5); (iii) $(83+17)(83-17) = 100 \times 66 = 6600$ (2).	1½+1½+2

Mistakes to avoid:

- Writing $(a+b)^2 = a^2 + b^2$; the middle term $2ab$ is missed.
- Writing $(a-b)^2 = a^2 - b^2$; correct is $a^2 - 2ab + b^2$, while $a^2 - b^2 = (a+b)(a-b)$.
- Multiplying only the first term of a bracket: $3(x+4) = 3x + 4$ instead of $3x + 12$.
- Sign errors when multiplying: $-2 \times (-3x) = +6x$ and $(x-3)(x-2)$ has $+6$ as the constant term.
- Adding unlike terms: $2x + 3x^2 = 5x^3$; x and x^2 are not like terms.
- Squaring a product without squaring the coefficient: $(3x)^2 = 9x^2$, not $3x^2$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 100	$(6+4)^2$
2	**(a)** 90	$(6+3)(6+4)$
3	**(d)** $12x$	The x^2 and constant terms cancel
4	**(b)** $4x+16$	Expand each bracket and collect like terms
5	**(a)** $6x^2+17x+5$	Multiply term by term and collect like terms
6	**(a)** $16x^2-40xy+25y^2$	$(a-b)^2=a^2-2ab+b^2$
7	**(c)** 8200	$(91+9)(91-9)=100\times82$
8	**(a)** 169	$(8+5)^2$
9	**(d)** 66	$(5+1)(5+6)$
10	**(d)** $20x$	The x^2 and constant terms cancel

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$57\times63=(60)^2-(3)^2$ (1) $=3600-9=3591$ (1).	1+1
12	$7x+35-2x+12$ (1) $=5x+47$ (1).	1+1
13	$(4x)^2+2(4x)(3y)+(3y)^2$ (1) $=16x^2+24xy+9y^2$ (1).	1+1
14	$24\times36=(30)^2-(6)^2$ (1) $=900-36=864$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$10x^2-2x+25x-5$ (2); $=10x^2+23x-5$ (1).	2+1
16	$299=300-1$ (1); $(300-1)^2=90000-600+1$ (1); $=89401$ (1).	1+1+1
17	LHS $=144+36=180$ (1.5); RHS $=2(81+9)=180$ (1.5).	1½+1½
	OR: $10x^2-20x+6x-12$ (2); $=10x^2-14x-12$ (1).	2+1
18	$52=50+2$ (1); $(50+2)^2=2500+200+4$ (1); $=2704$ (1).	1+1+1
	OR: LHS $=121+1=122$ (1.5); RHS $=2(36+25)=122$ (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Plot $=(5x+3)^2=25x^2+30x+9$ (2); pond $=9x^2$ (1); remaining $=16x^2+30x+9$ (1); at $x=10$: 1909 m² (1).	2+1+1+1
	OR: (i) $205^2=42025$ (1.5); (ii) $47\times53=2491$ (1.5); (iii) $(65+35)(65-35)=100\times30=3000$ (2).	1½+1½+2
20	Plot $=(2x+7)^2=4x^2+28x+49$ (2); pond $=9x^2$ (1); remaining $=-5x^2+28x+49$ (1); at $x=10$: -171 m² (1).	2+1+1+1
	OR: (i) $302^2=91204$ (1.5); (ii) $74\times66=4884$ (1.5); (iii) $(91+19)(91-19)=110\times72=7920$ (2).	1½+1½+2

Mistakes to avoid:

- Writing $(a+b)^2=a^2+b^2$; the middle term $2ab$ is missed.
- Writing $(a-b)^2=a^2-b^2$; correct is $a^2-2ab+b^2$, while $a^2-b^2=(a+b)(a-b)$.
- Multiplying only the first term of a bracket: $3(x+4)=3x+4$ instead of $3x+12$.
- Sign errors when multiplying: $-2\times(-3x)=+6x$ and $(x-3)(x-2)$ has $+6$ as the constant term.
- Adding unlike terms: $2x+3x^2=5x^3$; x and x^2 are not like terms.
- Squaring a product without squaring the coefficient: $(3x)^2=9x^2$, not $3x^2$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $3x + 29$	Expand each bracket and collect like terms
2	**(b)** $10x^2 + 26x + 12$	Multiply term by term and collect like terms
3	**(d)** $25x^2 - 20xy + 4y^2$	$(a-b)^2 = a^2 - 2ab + b^2$
4	**(b)** 1000	$(55+45)(55-45) = 100 \times 10$
5	**(d)** 121	$(9+2)^2$
6	**(a)** 54	$(2+4)(2+7)$
7	**(c)** $28x$	The x^2 and constant terms cancel
8	**(b)** $3x + 27$	Expand each bracket and collect like terms
9	**(b)** $12x^2 + 25x + 12$	Multiply term by term and collect like terms
10	**(c)** $9x^2 - 24xy + 16y^2$	$(a-b)^2 = a^2 - 2ab + b^2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$8x + 8 - 3x + 12$ (1) $= 5x + 20$ (1).	1+1
12	$(5x)^2 + 2(5x)(4y) + (4y)^2$ (1) $= 25x^2 + 40xy + 16y^2$ (1).	1+1
13	$82 \times 98 = (90)^2 - (8)^2$ (1) $= 8100 - 64 = 8036$ (1).	1+1
14	$6x + 24 - x + 5$ (1) $= 5x + 29$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$12x^2 - 9x + 16x - 12$ (2); $= 12x^2 + 7x - 12$ (1).	2+1
16	$98 = 100 - 2$ (1); $(100-2)^2 = 10000 - 400 + 4$ (1); $= 9604$ (1).	1+1+1
17	LHS $= 81 + 49 = 130$ (1.5); RHS $= 2(64+1) = 130$ (1.5).	1½+1½
	OR: $12x^2 - 18x + 2x - 3$ (2); $= 12x^2 - 16x - 3$ (1).	2+1
18	$203 = 200 + 3$ (1); $(200+3)^2 = 40000 + 1200 + 9$ (1); $= 41209$ (1).	1+1+1
	OR: LHS $= 289 + 9 = 298$ (1.5); RHS $= 2(100+49) = 298$ (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Plot $= (6x+1)^2 = 36x^2 + 12x + 1$ (2); pond $= 16x^2$ (1); remaining $= 20x^2 + 12x + 1$ (1); at $x = 10$: 2121 m^2 (1).	2+1+1+1
	OR: (i) $99^2 = 9801$ (1.5); (ii) $38 \times 42 = 1596$ (1.5); (iii) $(57+43)(57-43) = 100 \times 14 = 1400$ (2).	1½+1½+2
20	Plot $= (5x+4)^2 = 25x^2 + 40x + 16$ (2); pond $= 4x^2$ (1); remaining $= 21x^2 + 40x + 16$ (1); at $x = 10$: 2516 m^2 (1).	2+1+1+1
	OR: (i) $1004^2 = 1008016$ (1.5); (ii) $95 \times 105 = 9975$ (1.5); (iii) $(72+28)(72-28) = 100 \times 44 = 4400$ (2).	1½+1½+2

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- Multiplying only the first term of a bracket: $3(x+4) = 3x + 4$ instead of $3x + 12$.
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